

RESEARCH INTERESTS

Programming Languages, Parallel Computing, Formal Methods, and Software Engineering

EDUCATION

Carnegie Mellon University

Pittsburgh, USA

Ph.D. in Software Engineering (Advisor: Prof. Umut Acar)

Aug 2023 — Present

Pohang University of Science and Technology (POSTECH)

Pohang, Korea

Bachelor of Computer Science and Engineering/Mathematics (double major)

Feb 2018 — Feb 2023

- GPA : 4.26/4.3 (ranked 1st in class)
- Entered with *Program for highly Dedicated Students (PHDs)*, presidential scholarship and supporting program offered to one outstanding high school student in the year
- Graduated within 6 semesters (leave of absence for military duty during Jan 2020 — Dec 2021)

Korea Science Academy of KAIST (*High School for Science-Gifted Students*)

Busan, Korea

Graduated

Feb 2015 — Feb 2018

- GPA : 4.23/4.3 (ranked 1st in class)
- Funded by Korea Science Academy Fund Scholarship

RESEARCH EXPERIENCES

POSTECH Software Verification Laboratory

Pohang, Korea

Research Intern - Advised by Prof. Kyungmin Bae

Feb 2022 — May 2023

- **Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search**
 - Designed and implemented search heuristic for narrowing-based symbolic reachability analysis of distributed systems
 - Paper accepted to *International Workshop on Formal Techniques for Safety-Critical Systems 2022*

POSTECH Programming Languages Laboratory

Pohang, Korea

Research Intern - Advised by Prof. Sungwoo Park

June 2018 — June 2019

- **MR3**: General purpose distributed execution framework written in Scala
 - Designed and implemented locality-aware task scheduler achieving higher throughput

WORK EXPERIENCES

Kodebox

Seoul, Korea

Software engineer as military service alternative

Jan 2020 — Dec 2021

- **CodeChain Foundry**: Open-source blockchain engine based on composable module system
 - Worked as software engineer
 - Implemented BLS signature aggregation scheme to optimize block seal size
 - Designed interface for modules to communicate with consensus engine
 - CodeChain Foundry is written in Rust
- **Zuzu**: Web-based captable management software
 - Worked as full-stack software engineer
 - Main features implemented include regression testing framework for document generation service, payment, and more
 - Tech Stacks: Django (Python), React (TypeScript), GraphQL, Kubernetes, Google Cloud Platform

PUBLICATION

Narrowing and Heuristic Search for Symbolic Reachability Analysis of Concurrent Object-Oriented Systems, Byeongjee Kang and Kyungmin Bae, Science of Computer Programming 2024 (doi.org/10.1016/j.scico.2024.103097)

Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search, Byeongjee Kang and Kyungmin Bae, International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS) 2022 (doi.acm.org?doi=3563822.3568017)

HONORS AND SCHOLARSHIPS

Program for Highly Dedicated Students (POSTECH) *Feb 2018 — Feb 2023*

- Presidential scholarship and supporting program in POSTECH offered to **one outstanding freshman** each year

Presidential Science Scholarship (Ministry of Education of Korea) *Feb 2018 — Feb 2023*

- National science scholarship offered to top freshmen students in the field of science and technology

Global Leadership Program (CSE Department, POSTECH) *Sep 2019 — Present*

- Scholarship and research support program for selected students in computer science department

Domun Scholarship (POSTECH) *Feb 2019*

- Scholarship for freshmen with highest GPA

TEACHING EXPERIENCES

Student Mentoring Program (CSE Department, POSTECH) *Feb 2022 — June 2022*

- Served as tutor to teach Algorithm course

Intensive Research Program (POSTECH) *Jan 2019*

- Served as mentor in research experience program for science high school students

Voluntary Teaching (POSTECH Volunteering Club) *Jan 2019 — Dec 2019, Sep 2022 — Dec 2022*

- Served as voluntary teacher of math for unprivileged middle school students

EXCHANGE STUDENT & LANGUAGE PROGRAM

Summer Session Program TU Berlin
Data science course *Aug 2019 — Sep 2019*

English Program University of New South Wales
English course *Dec 2017 — Jan 2018*

AwesomeMath Summer Program Cornell University
Mathematics problem solving program for gifted middle and high school students *June 2016 — July 2016*

EXTRACURRICULAR ACTIVITIES & MISCELLANEOUS

POSTECH Decentralized Autonomous Organization (PDAO) POSTECH
Student association for blockchain development and research *Feb 2022 — Present*

- Served as mentor in blockchain core development team
- Designed consensus algorithm to be used in PDAO

Intensive Research Program POSTECH
Research experience program for science high school students - advised by Prof. Jae-Joon Kim *July 2017*

- Learned basic deep learning concepts and researched on techniques to reduce size of DNN

Research and Education (R&E) Program Korea Science Academy of KAIST
Group research program for juniors - Advised by Dr. Youngchul Jeon *Mar 2016 — Dec 2016*

- Researched properties of *Abelian Ring*, a field of noncommutative ring theory
- Presented result as school representative in Japan Super Science Fair (JSSF) 2016, at Ritsumeikan High School

Korea Young Physicists' Tournament (KYPT)

The Korean Physical Society

Team-based scientific competition for high school students

June 2017 — Jan 2018

- Designed and performed experiments, and debated with other teams on the outcome
- Won gold medal

POSCAT

POSTECH

Students club for algorithm study and competitive programming

Feb 2018 - Dec 2019

- Committee member of Postech Programming Contest 2019

POSTECH Student Ambassador

International Relations, POSTECH

Student-run POSTECH International PR group

Feb 2018 — Dec 2018

- Voluntary service to support exchange students visiting POSTECH

POBBA

POSTECH

Basketball club

Feb 2018 — Present

- Served as club secretary during the year 2019

LANGUAGES

Korean (native), English (fluent)

Last updated: Mar 8, 2023